

2020 AMC 10B Problem 5 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 5

Problem:

How many distinguishable arrangements are there of 1 brown tile, 1 purple tile, 2 green tiles, and 3 yellow tiles in a row from left to right? (Tiles of the same color are indistinguishable.)

Answer Choices:

- A. 210
- B. 420
- C. 630
- D. 840
- E. 1050

Solution:

Let's first find how many possibilities there would be if they were all distinguishable, then divide out the ones we overcounted.

There are $7!$ ways to order 7 objects. However, since there are $3! = 6$ ways to switch the yellow tiles around without changing anything (since they're indistinguishable) and $2! = 2$ ways to order the green tiles, we have to divide out these possibilities:

$$\frac{7!}{3! \cdot 2} = \boxed{(B)420}$$

OR

There are $\binom{7}{3}$ ways to arrange the yellow tiles which is 35. Then from the remaining tiles there are $\binom{4}{2} = 6$ ways to arrange the green tiles. And now from the remaining two tiles and two slots, we can see there are two ways to arrange the purple and brown tiles, giving us an answer of

$$35 \cdot 6 \cdot 2 = \boxed{(B)420}$$

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